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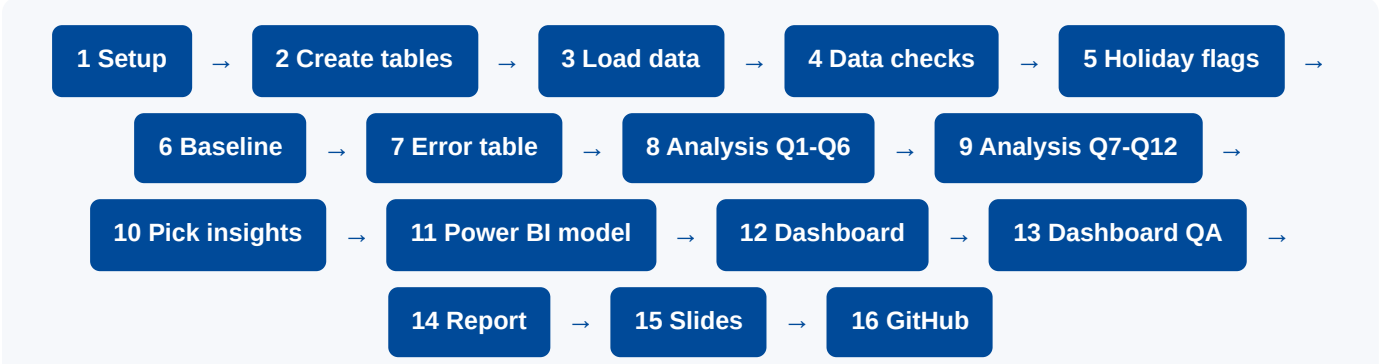
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Fast Track Steps and Flow

> **In simple words**
>
> This is your step-by-step plan. It has 16 steps in 5 phases. Every step has sub-steps, the result you should get, and a check to know you are done. Follow it in order.

How to use this document

> **The pattern (same for every step)**
>
> Every step has the same 6 rows: **Goal, Needs, Sub-steps, You get, Done when, If stuck.**
> 1) Read the Goal. 2) Do the sub-steps in order and tick each box. 3) Check **Done when**. 4) Only then go to the next step. Do not jump ahead. Each step feeds the next one.



Phase	Steps	Days	Main result
A. Setup	1 to 3	Day 1	Data inside PostgreSQL
B. Prepare data	4 to 7	Day 2 to 3	Error table (fact_error)
C. Analysis	8 to 10	Day 3 to 4	12 outputs, insight log, 6+ insights
D. Dashboard	11 to 13	Day 5 to 6	5-page Power BI dashboard
E. Communicate	14 to 16	Day 7	Report, slides, GitHub

Total time is about **46 hours**, which is 7 working days of 6 to 7 hours.

Daily routine (10 minutes at start, 10 at end)

- **Start:** open this document, find today's step, read its Goal and Done when.
- **End:** tick finished sub-steps. Save all files. Write 2 lines in your notes: what you did, what is next.
- **Stuck for more than 30 minutes?** Copy the exact error message, write what you tried, and ask for help. Do not skip the step.

Phase A: Setup (Day 1)

STEP 1: Folders and tools | Phase A | 1 hour	
Goal	Have a clean workspace and all tools installed.
Needs	Kaggle CSV files, a Windows PC.
Sub-steps	[ ] **1.1** Create the folder tree from Doc 00 (01_Datasets\raw, processed, 02_SQL_Scripts, 03_PowerBI, 04_Report, 05_Presentation, 06_Python_Optional, 07_Screenshots). [ ] **1.2** Put the 5 CSV files in `01_Datasets\raw`. Do not edit them. [ ] **1.3** Install PostgreSQL (with pgAdmin). Write down the password for user `postgres`. [ ] **1.4** Install Power BI Desktop (Microsoft Store or the Microsoft website). [ ] **1.5** Create an empty GitHub repository named `demand-forecasting-error-analysis`. [ ] **1.6** Open pgAdmin and connect to the server (right-click Servers, then connect).
You get	A working PostgreSQL connection and a clean folder.
Done when	pgAdmin shows your server connected, and Power BI Desktop opens.
If stuck	Wrong password: re-enter the one you set during install. Cannot find pgAdmin: search the Start menu.

STEP 2: Create the database and tables | Phase A | 1 hour	
Goal	Make empty tables that match the CSV files.
Needs	Step 1 done. Doc 03 (column meanings).
Sub-steps	[ ] **2.1** In pgAdmin: right-click Databases, Create, Database. Name it `demand_forecasting`. [ ] **2.2** Open the Query Tool on that database. [ ] **2.3** Open each CSV in Notepad and read the header line. Column names in SQL must match. [ ] **2.4** Write `01_create_tables.sql` with 5 tables: train, stores, holidays_events, oil, transactions. Types: date for dates, integer for ids, numeric for sales, text for names, boolean for transferred. [ ] **2.5** Run the script. Save it in `02_SQL_Scripts`.
You get	5 empty tables.
Done when	In pgAdmin, Schemas, Public, Tables shows all 5 tables.
If stuck	Type error: sales has decimals, so use numeric not integer. The oil file has a column named dcoilwtico.

STEP 3: Load the data | Phase A | 2 hours	
Goal	Fill the tables and prove that the data loaded correctly.
Needs	Step 2 done.
Sub-steps	[ ] **3.1** Right-click a table, Import/Export Data. Choose the CSV file, set Header = on, Delimiter = comma. [ ] **3.2** Load stores first (small), then holidays_events, oil, transactions, and last train (about 3 million rows, it takes a few minutes). [ ] **3.3** For oil: blank prices are normal (NULL). Keep them. [ ] **3.4** Run `SELECT COUNT(*)` on each table. Compare with the Kaggle page. Train should be about 3 million rows, stores = 54. [ ] **3.5** Write `02_load_data.sql` as a notes file with the row counts you got.
You get	5 loaded tables.
Done when	Row counts match. `SELECT * FROM train LIMIT 10` shows real values.
If stuck	Import fails on train: check the file path has no special characters, and that the column order matches the table.

Phase B: Prepare the data (Day 2 to 3)

STEP 4: Data checks | Phase B | 2 hours	
Goal	Find data problems before they spoil the results.
Needs	Step 3 done. Doc 03 section 7.
Sub-steps	[ ] **4.1** Count NULLs in sales, date, store_nbr and family. Expect none in train. [ ] **4.2** Check duplicates: group by store_nbr, family, date and look for count above 1. Expect none. [ ] **4.3** Check the date range (min and max date) and count distinct stores and families (54 and 33). [ ] **4.4** Find missing dates. Compare all calendar days with dates in train. 25 December is missing each year. That is expected. [ ] **4.5** Find each store's first sale date: `MIN(date)` where sales above 0, grouped by store. Save it as table `store_start`. [ ] **4.6** Find families with long zero-sales periods (share of zero rows per family). [ ] **4.7** Write `03_data_checks.sql` and a short notes file `data_issues.md`: issue, size, decision (exclude, flag, ignore).
You get	Issue list and the store_start table.
Done when	Every issue has a written decision.
If stuck	Too many zero rows: normal for new stores and rare families. Do not delete now, only note them.

STEP 5: Holiday flag table | Phase B | 3 hours	
Goal	Mark holidays for each store and date without double counting.
Needs	Step 4 done. Doc 03 section 4.
Sub-steps	[ ] **5.1** From holidays_events keep rows where `transferred = false` and type is Holiday, Transfer, Additional or Bridge. Ignore Event and Work Day. [ ] **5.2** Make a list of every store and every date in train (one row each). That is about 54 x 1,684 rows. [ ] **5.3** Set `is_holiday = 1` if a National holiday exists on that date, OR a Regional one where state = locale_name, OR a Local one where city = locale_name. Use EXISTS, so no duplicate rows appear. [ ] **5.4** Save as table `holiday_flag(store_nbr, date, is_holiday)`. Script: `04_holiday_flag_table.sql`. [ ] **5.5** Test: count rows = count of distinct (store_nbr, date). Christmas Eve is 1 for all stores. A Local holiday is 1 only for stores in that city.
You get	holiday_flag table.
Done when	No duplicate rows. Three hand tests pass.
If stuck	Row count too high: your join repeated rows. Use EXISTS instead of a plain JOIN.

STEP 6: Baseline forecast A and B | Phase B | 4 hours	
Goal	Create a simple forecast to compare with actual sales.
Needs	Steps 4 and 5 done. Doc 04 section 1.
Sub-steps	[ ] **6.1** Baseline A: for each store, family and date, take sales exactly 7 days earlier. Use `LAG(sales, 7)` and `LAG(date, 7)` with `PARTITION BY store_nbr, family ORDER BY date`. [ ] **6.2** Keep the row only if the date minus the lagged date = 7 days. This protects you from missing dates. [ ] **6.3** Baseline B: average of sales 7, 14, 21 and 28 days earlier. Use the same date check for all four. [ ] **6.4** Exclude rows before the store's first sale date (use store_start). The first 28 days of data have no baseline, so they drop out. [ ] **6.5** Save as view `baseline_view(store_nbr, family, date, actual, forecast_a, forecast_b)`. Script: `05_baseline_forecast.sql`. [ ] **6.6** Spot-check 5 rows by hand: forecast_a must equal the actual of the same store and family 7 days earlier.
You get	baseline_view.
Done when	5 spot-checks match. No NULL forecast_a.
If stuck	Forecast looks wrong near 25 December: that is the missing-date rule at work. Check the exact date difference.

STEP 7: Error table (the main table) | Phase B | 3 hours	
Goal	Build one table that every analysis and the dashboard will use.
Needs	Steps 5 and 6 done. Doc 04.
Sub-steps	[ ] **7.1** Join baseline_view, holiday_flag and stores (city, state, type, cluster). Add onpromotion from train. [ ] **7.2** Add `abs_error = ABS(actual - forecast_a)`, `excess_units = GREATEST(forecast_a - actual, 0)`, `shortfall_units = GREATEST(actual - forecast_a, 0)`. Add the same abs error for B. [ ] **7.3** Add flags: `is_payday` (day 15 or last day of the month), month, quarter, weekday, period label (Christmas, Carnival, school start, Mother's Day) with CASE WHEN, and a flag if the baseline day (7 days earlier) was a holiday. [ ] **7.4** Save as table `fact_error` (CREATE TABLE AS). Add indexes on family, date and store_nbr. Script: `06_error_table.sql`. [ ] **7.5** Validate: total sales equals the sum in baseline_view. Excess minus shortfall equals total forecast minus total actual. Calculate WAPE for one family by hand and compare.
You get	fact_error table.
Done when	All three validation checks pass.
If stuck	The identity check fails: look for rows where forecast is NULL, or double rows from a join.

Phase C: Analysis (Day 3 to 4)

> **Rule for steps 8 and 9**
>
> After **every** query, add one row to the insight log (Doc 12): finding with numbers, comparison, likely reason, action, confidence. Write the hypothesis **before** you run the query. A rejected hypothesis is also an insight.

STEP 8: Analysis Q1 to Q6 | Phase C | 5 hours	
Goal	Find which families, regions and months are hardest to predict.
Needs	Step 7 done. Doc 05 (questions), Doc 12 (hypotheses and log).
Sub-steps	[ ] **8.1** Q1 Overall health: WAPE, Accuracy, Bias %, excess and shortfall units, by year. Output `overall_health.csv`. [ ] **8.2** Q2 Hardest families: WAPE by family with a minimum volume rule, ranked. Output `family_error_summary.csv`. [ ] **8.3** Q3 Pareto: error share and cumulative share by family. Find the families that make 80% of error. Output `family_pareto.csv`. [ ] **8.4** Q4 Direction: excess vs shortfall by family and a label (Over, Under, Balanced). Output `family_direction.csv`. [ ] **8.5** Q5 Regions: WAPE and Bias % by state, city, store type, cluster. Output `region_performance.csv`. [ ] **8.6** Q6 Seasons: WAPE and Bias % by month, quarter and period label. Output `season_error.csv`. [ ] **8.7** Export each result: in pgAdmin run the query, then use the Download as CSV button. Save to `01_Datasets\processed`. [ ] **8.8** Save queries in `07_analysis_Q01_Q06.sql`. Fill 6 rows in the insight log.
You get	6 CSV files and 6 insight log rows.
Done when	Each output has a log row with a number and a comparison.
If stuck	WAPE above 100% for a family: it has very small sales. Check volume before you trust it.

STEP 9: Analysis Q7 to Q12 | Phase C | 4 hours	
Goal	Find when and why the baseline fails.
Needs	Step 8 done.
Sub-steps	[ ] **9.1** Q7 Holidays: WAPE and Bias % for holiday, week-after-holiday and regular days. Output `holiday_impact.csv`. [ ] **9.2** Q8 Promotions: compare onpromotion = 0 and above 0. Output `promotion_impact.csv`. [ ] **9.3** Q9 Paydays: 15th and last day vs other days. Output `payday_impact.csv`. [ ] **9.4** Q10 Weekdays: WAPE and Bias % by weekday. Output `weekday_error.csv`. [ ] **9.5** Q11 Baseline comparison: WAPE of A and B by family, and which is better. Output `baseline_compare.csv`. [ ] **9.6** Q12 Trend: WAPE by month for the top 5 families, with a 3-month rolling WAPE. Output `error_trend.csv`. [ ] **9.7** Save queries in `08_analysis_Q07_Q12.sql`. Fill 6 more rows in the insight log.
You get	6 more CSV files and an insight log with 12 rows.
Done when	All 12 questions have an output and a log row.
If stuck	Results look flat: that is a finding too. Write 'no clear difference' and the numbers.

STEP 10: Pick the best insights | Phase C | 2 hours	
Goal	Turn 12 results into 6 to 8 strong insights and clear actions.
Needs	Insight log filled. Doc 12 (card, sentence formula, recommendation types).
Sub-steps	[ ] **10.1** Mark the log rows that are big (high error share or many units) and clear. [ ] **10.2** Choose the 6 to 8 best. Make sure at least one covers family, one region, one season or event, and one direction (over or under). [ ] **10.3** Write one insight card for each (Doc 12 format). Use the sentence formula. [ ] **10.4** Write recommendations with owner and priority. Rank by impact and effort. [ ] **10.5** Pick the top 3 for the executive summary. Save everything in `04_Report\insight_cards`.
You get	6 to 8 insight cards and a recommendation list.
Done when	Every card has evidence, comparison, reason, action and a limit. No invented numbers.
If stuck	Not sure which insights are strong: choose the ones with the largest share of error and a clear action.

Phase D: Dashboard (Day 5 to 6)

STEP 11: Power BI connection, model and measures | Phase D | 3 hours	
Goal	Get the data into Power BI and make sure the numbers match SQL.
Needs	Steps 7 and 10 done. Doc 06 sections 2 and 6.
Sub-steps	[ ] **11.1** Home, Get Data, PostgreSQL database. Server: localhost. Database: demand_forecasting. Choose Import. If Power BI asks for the Npgsql driver, install it. [ ] **11.2** Load fact_error and stores. If it is too slow, aggregate in SQL to store x family x week and say so in the report. [ ] **11.3** In Power Query remove columns you do not need and set data types. Rename columns to readable names. [ ] **11.4** Build dim_date and dim_family. In Model view create one-to-many relationships: dimensions to fact. [ ] **11.5** Create a table called Measures. Add: Total Actual Sales, Total Forecast, Total Abs Error, WAPE %, Accuracy %, Bias %, Excess Units, Shortfall Units, Error Share %, High Error Flag, Direction Label. Use DIVIDE. [ ] **11.6** Add a card and a table. Compare Total Sales and WAPE with your SQL results.
You get	Working model with measures.
Done when	Total sales, abs error and WAPE match SQL.
If stuck	Numbers differ: check for rows excluded in SQL, filters on the page, or a wrong relationship direction.

STEP 12: Build the 5 pages | Phase D | 6 hours	
Goal	Build the story, one page at a time. Headline first.
Needs	Step 11 done. Doc 06 section 4.
Sub-steps	[ ] **12.1** Set up: 16:9 page, a blue-gray theme, the 4 slicers (date, region, family, store type), page navigation buttons, a Reset filters bookmark. [ ] **12.2** Page 1 Executive Summary: 5 KPI cards, bar of top 5 families, monthly WAPE line, takeaway text box. Write the headline sentence. [ ] **12.3** Page 2 Category Deep-Dive: scatter (sales vs WAPE), Pareto bar and line, stacked bar of excess vs shortfall. Add drill-through to a Family Detail page. [ ] **12.4** Page 3 Regional: map or bar by city, matrix state to city with colors, bar by store type. Add a tooltip. [ ] **12.5** Page 4 Seasons and Events: WAPE by month, holiday vs week-after vs regular, weekday x month heatmap, payday and promotion cards. [ ] **12.6** Page 5 Insights and Actions: your 5 insight cards as text boxes with finding, evidence, action, owner, priority. Add buttons to the proof pages. [ ] **12.7** Check every page: maximum 6 visuals, one message each, readable titles, units shown.
You get	A complete dashboard (.pbix) in `03_PowerBI`.
Done when	Each page has a headline that is true and a clear takeaway.
If stuck	A visual is crowded: remove it. Fewer charts are better.

STEP 13: Dashboard QA and screenshots | Phase D | 1.5 hours	
Goal	Be sure the dashboard is correct before anyone sees it.
Needs	Step 12 done. Doc 09 (QA list).
Sub-steps	[ ] **13.1** Click every slicer. Numbers must change correctly. [ ] **13.2** Compare Page 1 totals with the SQL outputs once more. [ ] **13.3** Read each headline with filters cleared. Fix any that are not true. [ ] **13.4** Look for overlapping or cut-off visuals in full screen. [ ] **13.5** Take one screenshot per page into `07_Screenshots`. Export the dashboard as PDF (File, Export).
You get	A checked dashboard and 5 screenshots.
Done when	All QA boxes for the dashboard are ticked.
If stuck	A slicer does nothing: check relationships and the Edit interactions setting.

Phase E: Communicate (Day 7)

STEP 14: Final report | Phase E | 4 hours	
Goal	Write an 8 to 10 page report a manager can read in 10 minutes.
Needs	Insight cards, screenshots. Doc 08.
Sub-steps	[ ] **14.1** Write the executive summary last, but put it first (template in Doc 13). [ ] **14.2** Write context, data and method (baseline, metrics, high-error rule, limits). [ ] **14.3** Write one findings section per insight card. Add one chart and the key numbers each time. [ ] **14.4** Write the recommendation table (action, owner, priority, evidence, how to measure). [ ] **14.5** Write limits and next steps. Add the appendix (metric definitions, SQL file list). [ ] **14.6** Read it once for simple words and once for invented numbers. Export to PDF into `04_Report`.
You get	`Final_Project_Report.pdf`.
Done when	Every finding has an action. Every cause says 'likely' unless proven.
If stuck	Too long: cut the weakest finding.

STEP 15: Executive summary and slides | Phase E | 3 hours	
Goal	Be ready to present the project clearly in 2 minutes and in 10 minutes.
Needs	Report done. Doc 13.
Sub-steps	[ ] **15.1** Fill the one-page executive summary template. [ ] **15.2** Make 8 slides with sentence headlines (title, why it matters, big picture, which families, which way, where and when, why likely, what to do). [ ] **15.3** Write the 2-minute story from Doc 14. Say it aloud 3 times. [ ] **15.4** Prepare answers to the 5 common questions in Doc 13. [ ] **15.5** Save slides in `05_Presentation`.
You get	Executive summary, 8 slides, and a practiced story.
Done when	You can explain the project in 2 minutes without reading.
If stuck	You forget the numbers: put 3 key numbers on a small card and keep it with you.

STEP 16: GitHub and resume | Phase E | 2 hours	
Goal	Make the project visible to recruiters.
Needs	Everything finished. Doc 13 (README outline), Doc 14 (resume lines).
Sub-steps	[ ] **16.1** Upload SQL scripts, data check notes, report PDF, slides, screenshots and the dashboard file. Do not upload the raw Kaggle files (large). Add a link to the Kaggle page. [ ] **16.2** Write README.md: problem, decisions, data and tools, method, top insights with screenshots, recommendations, limits, how to run the SQL. [ ] **16.3** Add 3 to 4 resume lines using your real numbers (Doc 14). [ ] **16.4** Update your LinkedIn headline and add the project link.
You get	A public GitHub repository and an updated resume.
Done when	A stranger can understand the project from the README in 2 minutes.
If stuck	Large file error on GitHub: skip the raw data and share the link instead.

If you are short of time: minimum version

- Baseline A only. Analysis Q1 to Q6. Dashboard Pages 1, 2 and 5.
- Report with 4 findings. Slides: summary, 3 findings, actions.
- **Never skip:** step 4 data checks, step 5 holiday flags, matching totals (step 11), recommendations and README.

Progress tracker

Step	Name	Status (To do / Doing / Done)	Date finished	Notes
1	Folders and tools			
2	Create tables			
3	Load data			
4	Data checks			
5	Holiday flag table			
6	Baseline A and B			
7	Error table			
8	Analysis Q1-Q6			
9	Analysis Q7-Q12			
10	Pick insights			
11	Power BI model			
12	Build 5 pages			
13	Dashboard QA			
14	Final report			
15	Slides			
16	GitHub and resume			

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